**Department of Electrical and Computer Engineering**

**North South University**

# Senior Design Project

# Diffusion Models for Image Inpainting

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**Assistant Professor**

**ECE Department**

**Fall 2025**

# LETTER OF TRANSMITTAL

December,2025

To

Prof. Mohammad Abdul Matin

Chairman,

Department of Electrical and Computer Engineering

North South University, Dhaka

Subject: Submission of Capstone Project Report on “Diffusion Models for Image Inpainting.”

Dear Sir,

With due respect and humble submission, we would like to submit our Capstone Report on “**Diffusion Models for Image Inpainting**: A Comparative Evaluation Using **DDPM, Stable Diffusion, and Repaint** as part of our BSc program. The report deals with a Multi-Level Based Home Security System. As this project really helped us to gain practical hands-on experience from the practical field and implement it in our real life so this project has become very significant to us. We tried our level best to fulfill all the dimensions that are required for this report to meet the maximum competence.

We would be highly grateful and obliged if you kindly accept our report and give us your valuable judgment. It would be our immense pleasure if you find our report effective, impactful, and informative to have a clear perspective on the issue.

Sincerely,

Kazi Tazrian Mon

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# APPROVAL

Kazi Tazrian Mon (2132798642), Nur Ibne Kawsar Zitu (2212021042), and Md Minhajul Islam (2212021042) from the Department of Computer Science & Engineering at North South University, Bangladesh (NSU), have completed the capstone project titled:
"**Diffusion Models for Image Inpainting**: A Comparative Study Using DDPM, Stable Diffusion, and Repaint". The project was carried out under the supervision of M. Shifat-E-Rabbi, in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering (CSE). This report has been examined and is hereby accepted as satisfactory.

**Supervisor's Signature**

**M. Shifat-E-Rabbi**
**Assistant Professor**
Department of Electrical and Computer Engineering
North South University
Dhaka, Bangladesh.

**Chairman's Signature**

**Mohammad Abdul Matin**
**Professor**
Department of Electrical and Computer Engineering
North South University

Dhaka, Bangladesh.

# DECLARATION

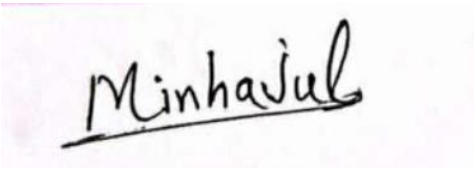
We hereby declare that the project titled “Diffusion Models for Image Inpainting: A Comparative Study Using DDPM, Stable Diffusion, and RePaint” is entirely the result of our own work. No part of this project has been submitted, either partially or fully, for the award of any other degree or diploma at any academic institution. All information, results, and findings included in this report are based on our research and implementation. Any prior work or external source referenced in this report has been properly acknowledged and cited according to academic standards. We also affirm that we have maintained the plagiarism policy as instructed by our project supervisor. All project-related materials shall remain confidential and will not be disclosed without the formal consent of the supervisor.

Student’s names & signatures

1. **Kazi Tazrian Mon**

2. **Nur Ibne Kawsar Zitu**

3. **Md Minhajul Islam**



# ACKOWLEDGEMENTS

We would like to sincerely express our immense gratitude to our respected project supervisor, M. Shifat-E-Rabbi, Department of Electrical & Computer Engineering, for his consistent guidance, impactful feedback, judgment, and, most importantly, his continuous support throughout the development and implementation of our capstone project. His proficiency and extensive experience in Machine Learning and Computer vision have significantly enhanced our understanding of numerous diffusion models and their implementation, which has largely strengthened the caliber of our research and machine learning model implementations.

We are grateful to the **Department of Electrical & Computer Science Engineering, North South University, Dhaka, Bangladesh**, as well as for providing the appropriate academic environment, resources regarding computations, and engagement essential to complete this project successfully.

We also would like to extend our appreciation to our classmates and friends for their effective discussions, inspiring and utmost assistance whenever we were in need. Last but not least, we sincerely owe heartfelt gratitude to our families for their unwavering support, patience, and continuous encouragement during the entire semester of this project.

# ABSTRACT

## CSE499A Capstone Project “Diffusion Models for Image Inpainting”

This project illustrates and explores the usage of diffusion-based generative models for the restoration of missing regions in images as a task of image inpainting. Currently, diffusion models have become one of the impactful and effective approaches for the generation of high-quality images, thus their comparative evaluations across multiple different architectures have more likely remained as a core topic of study. For the development of a clearer and better understanding of both the forward and backward propagation diffusion methods, a fundamental diffusion model was first implemented and then trained from the basics based on a standard dataset in this work. To evaluate and analyze the effectiveness of modern diffusion processes, a pretrained high-resolution-based inpainting model was implemented on a large-scale face dataset. A lightweight supervised model was also architectured well to deeply notice the differences between generative restoration models and traditional image-to-image learning. Lastly, a specialized inpainted diffusion model was evaluated to benchmark architectural consistency and quality restoration. The final implementation shows the experimental results, which describe that the pretrained generative diffusion models generate the most coherent and realistic outputs where on the other hand the self-trained model illustrates the fundamental learning traits of diffusion

methods. The comparative studies focus on the strengths, boundaries, and real-life significance of diffusion models for image restoration works.

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# Chapter 1 Introduction

## 1.1 Background and Motivation

Image inpainting has become one of the main problems in computer vision; it represents the reconstruction of missing or corrupted areas or territories of an image in a visually coherent approach. This work was classically approached using interpolation, patch-based synthesis, or conventional convolutional neural network methods. Though these earlier models were able to restore simple textures or small missing regions, they often failed in recovering small semantic forms like faces, objects, or any sort of complex backgrounds. Because of these shortcomings, their practical effectiveness in real-world applications remained confined.

Image restoration tasks have finally developed due to the emergence of generative models, especially advanced diffusion models. Diffusion models function through two gradual processes, which are noising and denoising, knowing about the underlying data distribution, and validating highly realistic image synthesis. Diffusion Models such as DDPM (Denoising Probabilistic Model), Denoising Implicit Models, and the more advanced latent diffusion frameworks have shown groundbreaking capabilities in works that include the generation of images, editing, and inpainting. These models' impressive qualities have surpassed the

preceding generative approaches in terms of authenticity, formation preservation, and availability

Inpainting holds significant applications in photographic advancement, medical imaging, face editing, unwanted object removal, and historical archives restoration. The demand for High-quality restoration methods is also growing across industries such as content creation, digital forensics, entertainment, and augmented reality. Manual editing workloads can be reduced, and automated reconstruction processes can also be enabled using a reliable inpainting system.

The motivation behind demonstrating this project comes from the understanding and comparison of the potential of various diffusion-based models for image inpainting. Basically, pre-trained models like Stable Diffusion Model and Repaint offer state-of-the-art performance, so establishing and training a diffusion model from the ground level provides valuable insight into how these systems learn and function internally. Through validating a traditional DDPM, a lightweight supervised U-Net, and two advanced pretrained diffusion models on the same dataset and masking strategies from the original images, our project's purpose is to focus on the reinforcement, limitations, and practical trade-offs between fundamental and advanced methods.

The comparative study of this project is similarly significant as diffusion models' continuation influence on real-world applications. Ultimately, this project forms a bridge between theoretical understanding and practical evaluation to contribute meaningfully to insights into the fast-evolving aspect of generative restoration of images

## 1.2 Purpose and Goal of the Project

By analyzing and comparing the effectiveness of various diffusion-based generative models for the task of image inpainting is the main aim of this project. While the diffusion models are on a loop of redefining the landscape of image synthesis and image restoration, understanding their practical strengths and boundaries, a formed and structured validation of their performance under specific

conditions is needed. This project creates a bridge between theoretical understanding with practical evaluation, contributing to meaningful insights into the fast-evolving aspect of generative image restoration and benchmarking a range of diffusion architectures.

The primary goal of this project is the implementation of a DDPM (Denoising Probabilistic Model) from scratch to develop the core understanding of both the forward and reverse denoising techniques that stimulate modern diffusion models. Diffusion models learn data distribution and missing content restoration by manually building the model, training it on the CIFAR-10 dataset, and applying it to a basic inpainting task, and thus offer significant insights.

Evaluating pretrained state-of-the-art diffusion inpainting models, particularly Stable Diffusion Inpainting and Repaint, is the second goal of this project. These models are operated on high-resolution CelebA-HQ images and illustrate more advanced semantic reasoning- characteristics that traditional diffusion models trained from the ground level cannot gain at small scales. The project offers an equal comparison of their capabilities by applying similar masking conditions to these models.

The third goal of this project is to develop a lightweight U-Net supervised inpainting model that is masked on CelebA-HQ images. This provides a practical baseline for noticing and analyzing how non-generative convolutional designs perform, linked to diffusion-based methods. It focuses on the difference between deterministic supervised learning and generative restoration.

The notion of our project consists of its comprehensive and multi-level comparison, such as :

- Implemented diffusion model from scratch
- A custom supervised U-Net model
- Pretrained diffusion inpainting models of two state-of-the-art.

All have been evaluated using the same dataset, same mask types, as well as the same evaluation metrics(SSIM and PSNR).

So overall, the aim of this project is not only to focus on performing inpainting but also to build a complete learning to application pipeline: implementation of

foundational models, applying modern pretrained models, building a supervised baseline, and lastly comparing all of them through qualitative and quantitative analysis.

## 1.3 Organization of the Report

The report of this project is structured into eight different chapters, each addressing a particular component of this project in a logical and appropriate approach.

**Chapter-1: Introduction**

In this chapter, the motivation and the foundation behind implementing the models are given. It mentions the objectives, goals proposed, scope, contributions, and challenges related to diffusion-based image inpainting. It illustrates the overall architecture of the report as well.

**Chapter 2: Literature Review**

This chapter illustrates a detailed review of the existing literature papers on diffusion models, inpainting methods, as well as generative architectures. It discusses the traditional DDPM frameworks, DDIM sampling, U-Net variations, Stable diffusion, and Repaint by establishing the theoretical context on which based on the project is based.

**Chapter 3: Methodology**

This chapter explains the full workflow of the project, from the dataset preparation and mask generation to the realization of the DDPM model from ground level, up to integrating the pre-trained Stable Diffusion and Repaint pipelines, and thus creates a supervised U-Net inpainting system. It gives overall and detailed information on the training methods, architectural decisions, and chosen metrics for evaluation.

**Chapter 4: Investigation, Experiments, Results, and Discussion**

This chapter illustrates the experimental setup followed by both the quantitative and qualitative evaluations for all methods tested. In this work, the performance of

DDPM, a custom U-Net, Stable Diffusion, and Repaint is analyzed using PSNR, SSIM, and visual comparisons. Multiple strengths, weak points, and traits have been observed for each model when faced with different masking conditions are discussed.

**Chapter 5: Impacts of the Project**

This chapter emphasizes the technical, societal, and academic impacts of the project, which describes and discusses how diffusion models contribute to various segments like digital restoration, medical imaging, film editing, and AI assistance of content generation.

**Chapter 6: Project Planning and Budget**

In this chapter, the timeline, computational requirements, task distribution, and resource allocation have been described. Also, the chapter includes the usage of GPU, cloud computing needs, and associated constraints, as the project is computationally intensive.

**Chapter 7: Complex Engineering Problems and Activities**

This chapter describes how the work has been performed in this project to meet all the complex engineering problem characteristics. Analyzing problems, modeling, and considering designs, various uses of advanced tools, groupwork, ethics, communication, and learning components have been examined in this chapter.

**Chapter 8: Conclusions**

The final chapter overall summarizes the fundamental findings of this project, compares all of the models' performance, also includes recommendations for future improvements, such as higher-capacity DDPMs, better masking strategies, and extra pretrained diffusion models.

# Chapter 2 Research Literature Review

## 2.1 Existing Research and Limitations

In image inpainting, recent advancements have been driven mostly by generative diffusion-based models, latent diffusion models, and multimodal editing frameworks. In order to develop semantic consistency, structural coherence, and user control, several state-of-the-art approaches have been put forward. Each methods exhibit clear limitations in spite of its significant progress. For motivating and inspiring a comprehensive comparative research, understanding their strengths and weaknesses is very necessary, as presented in the project.

Biswas et al. proposed PIXELS, which is a latent diffusion–based method that enables pixel-wise and region-level editing through example images. In order to maintain consistency during reconstruction in his method it uses low-dimensional latent representations and progressive editing algorithms. PIXELS exceeded performance in multiple strong baselines that include Versatile Diffusion, Paint-by-Example, IP Adapter, and MimicBrush, leading to achieving the CLIP-I and FID scores. On the contrary, the presented approach lacks an automated segmentation pipeline, making it a bit less user-friendly; thus, manual intervention is necessary for mask preparation. This also focuses on a point that is more general to many inpainting systems: they often depend on high-quality segmentation inputs**[2].**

Buburuzan et al. introduced MObI, a camera and LiDAR fusion approach for multimodal object inpainting using 3D bounding boxes to restore geometric consistency. While the model enhances realism and spatial accuracy, it suffers from degraded performance in out-of-domain scenes and sometimes causes unintended changes to background regions. This points out a common restriction across the different models on diffusion inpainting: their susceptibility to domain shifts and inability to generalize to other distributions than those they were trained on**[1].**

Kim et at proposed Region-Aware Diffusion Models(RAD), which focus only on masked areas during denoising by modifying the noise scheduling mechanism. RAD gains high-quality inpainting results at a significant rate at a faster speed, up to 100x faster than Repaint. RAD still heavily relies on mask type and needs retraining or fine-tuning under different masking scenarios despite its efficiency.

This results in a limitation in many diffusion-based models in balancing flexibility and availability, with training consistency remaining difficult**[4].**

Lugmayr et al.'s Repaint, which he proposed, is one of the most impactful and inspiring inpainting diffusion models. Repaint leverages a pretrained unconditional DDPM and implements a resampling strategy to bridge semantic gaps within the masked regions. It shows the best performance for a variety of complex mask types, outperforming models like AOT, DSI, and ICT. Moreover, Repaint can be rather computationally expensive since many forward–backward diffusion cycles are required; hence, it is slower compared to modern alternatives. Performance also often varies depending on mask size, with thick-mask scenarios sometimes showing better results with other models, such as Lama**[8].**

A developed Uni-paint proposed by Yang et at is a multimodal inpainting framework which supports unconditional, text-driven, stroke- driven and exemplar-driven assistance. Uni Paint focuses on several limitations of single-modal methods by providing a wide range of controls and also by building on top of Stable Diffusion. This method achieves groundbreaking quantitative results across many tasks. Still Uni paint model heavily depends on masked attention mechanisms and fine-tuning that can limit scalability while adapting to new domains or higher resolution datasets**[6].**

Wasserman et al. proposed Paint-by-Inpaint, an inverse inpainting strategy, where models learn to add objects by learning how to remove them, showing for the first time state-of-the-art performance for object insertion. Despite its impressive results, this method relies on an extremely large dataset (PIPE) of one million image pairs and, therefore, requires computationally expensive and resource-intensive training. Hence, the methods that relies on such large-scale data have limited accessibility for smaller research groups or applications require portability**[7].**

**Summary of Limitations:**

After surveying various research papers, the following consistent limitations arise and these are High computational cost for diffusion-based inpainting: Repaint, PIPE-based models. The domain sensitivity that results in poor performance on unseen distributions: MObI, Uni-paint. Dependent upon high-quality masks, hence limiting usability in real-world scenarios: PIXELS, RAD. This Complex training pipelines that involve large datasets or multimodal supervision. Its Speed–quality trade-offs: while faster models typically sacrifice detail, RAD, high-quality models run slower, Repaint.

# Chapter 3 Methodology

## 3.1 System Design

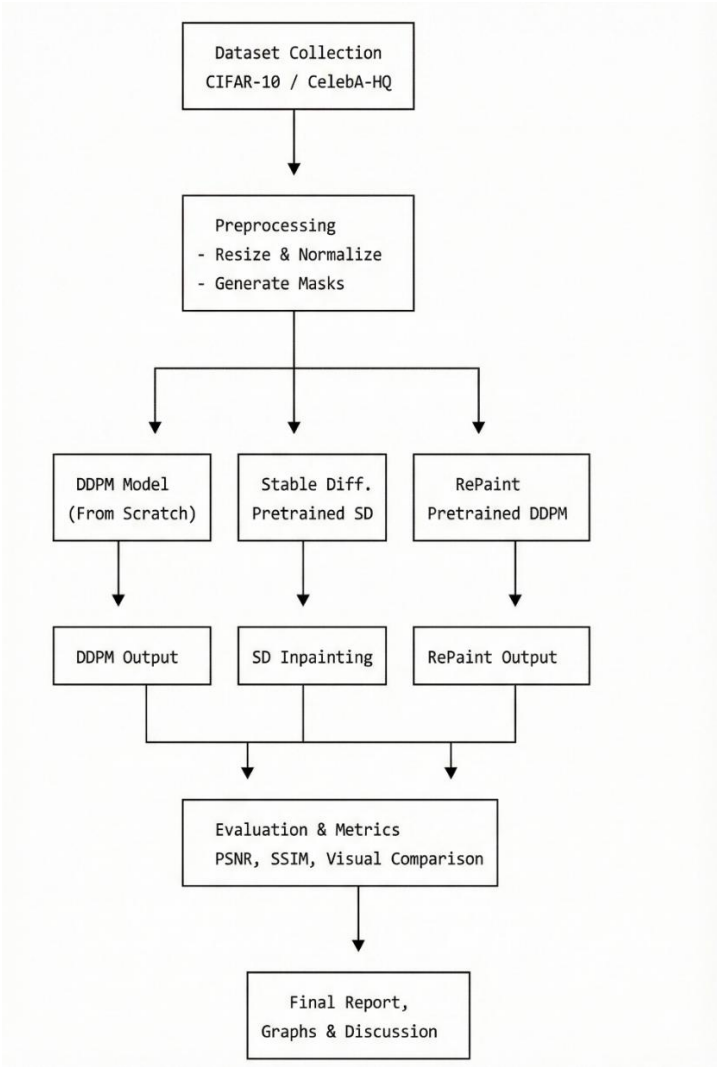
Our system design of this project is done in four interconnected modules, each and responsible for a different kind of inpainting pipeline. A high-level diagram for this system. The data acquisition and preprocessing module is: It loads CIFAR-10 (Krizhevsky, n.d.)**[5]** and CelebA-HQ (Karras, n.d.)**[3]** datasets. It also normalizes the image and generates masks. It masks in the center and a random square-sized box. It prepares supervised trained pairs for the U-net model.

Model development module: The DDPM implementation id forward noising schedule, u net architecture, training loop, and DDPM&DDIM sampling. Stable diffusion inpainting pretrained latent diffusion model for inpainting the image. The custom U-net supervised learning-based image-to-image network for inpainting the network. The RePaint inpainting model uses forward reverse jumps in the diffusion resampling inpainting method. The evaluation and benchmarking module computes PSNR and SSIM inside the masked part. It generates qualitative visual comparisons. It also creates a table and graph to compare model performance. The result produces

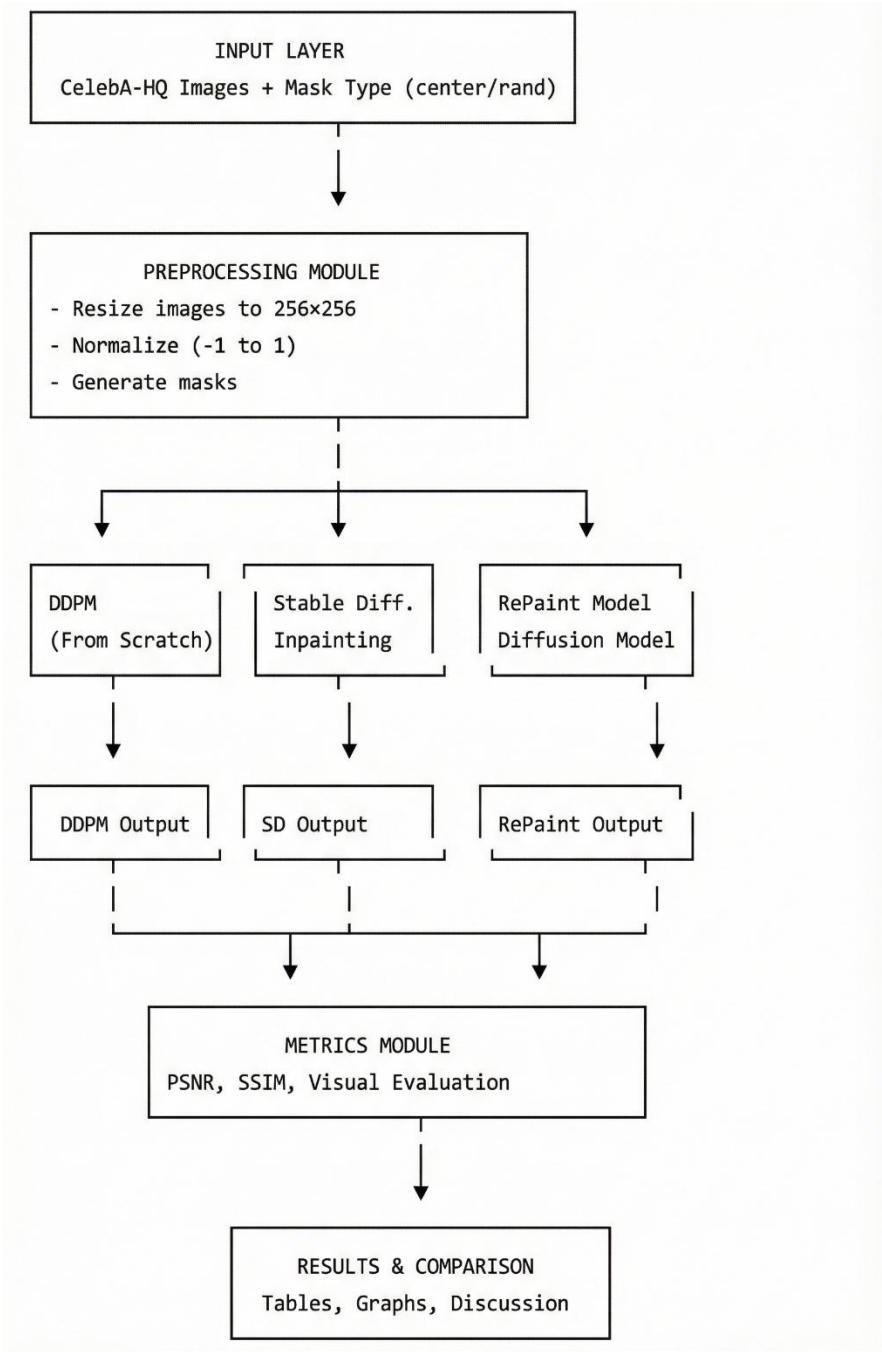
side-by-side outputs. It saves the outputs in Google Drive. It generates metrics summary and comparison charts.

The modular design ensures that every model used in project DDPM, U-Net, Stable Diffusion, and RePaint can run independently and share common preprocessing and evaluation pipelines. It's designed for reproducibility and support re-running inference even after a runtime reset, and storing output and checkpoints saved in Google Drive.

**System Design Diagram**



## System Architecture Diagram



This modular architecture ensures that each model-DDPM, U-Net, Stable Diffusion, and Repaint-can be executed independently with shared preprocessing and evaluation pipelines. The system is designed for reproducibility and supports re-running inference even in the case of runtime resets by storing outputs and checkpoints in Google Drive.

## 3.2 Hardware and Software Components

Our project highlights deep learning based inpainting on images, which requires both software and hardware frameworks and hardware accelerators for model training and inference, which is regarded as output.

A detailed explanation of the components that are used in this project is mentioned below:

We use Python 3.10. It is a primary programming language for deep learning. PyTorch is used to implement DDPM, U-net, and all training loops. We use the diffuser library to provide a pretrained Stable Diffusion and RePaint model. It is used for training DDPM and U-Net. For PSNR and SSIIM evaluation metrics, we use scikit-image. For visual results, masking and comparison output, we use matplotlib. We use Google Drive for storage of datasets, trained models, masks, and outputs.

TABLE I. A SAMPLE SOFTWARE/HARDWARE TOOLS TABLE

| **Tool** | **Functions** | **Other similar Tools (if any)** | **Why selected this tool** |
|---|---|---|---|

| | | | |
|---|---|---|---|
| Python 3.10 | Main programming language for ML development | Julia, MATLAB | Widely used in AI research; strong library support |
| PyTorch | Implementing DDPM, U-Net, training loops | TensorFlow | Easy to customize models; industry-standard |
| Diffusers Library | Pretrained Stable Diffusion & RePaint pipelines | Hugging Face Transformers, OpenAI Glide | Stable API for diffusion models; supports many architectures |
| Google Colab GPU (Tesla T4 / P100) | Accelerates training, sampling, and inference | Kaggle GPU, AWS EC2, Paperspace | Free/affordable GPUs; built-in CUDA; no local setup required |
| Scikit-Image | PSNR & SSIM metrics | OpenCV | Simple and reliable evaluation functions |
| Matplotlib | Visualization (images, graphs, comparisons) | Seaborn, Plotly | Easy to integrate with notebooks |
| Google Drive | Stores datasets, checkpoints, trained models, outputs | Dropbox, OneDrive | Persistent storage for large files and reproducibility |

## 3.3 Hardware and Software Implementation

This section describes how the software modules were implemented and executed. The implementation started with the preparation of the dataset, whereas the CIFAR-10 dataset was directly loaded from Torchvision, supporting DDPM training. CelebA-HQ images were stored on Google Drive in parallel and would be used for the experiments on Stable Diffusion, U-Net, and RePaint. It also programmatically generated masks in order to facilitate inpainting tasks, both center square masks and random rectangular masks.

For the implementation of DDPM (Part A), by using a linear beat schedule, the forward propagation was carried out. A custom U-Net structure was architectured with timestep embeddings, and the model was trained for ten epochs. The Reverse sampling propagation was also implemented using both DDPM and DDIM methods.

In the Stable Diffusion inpainting stage (Part B), first, the pre-trained model "runwayml/stable-diffusion-inpainting" was loaded and then applied to CelebA-HQ images with a resolution of 256×256. Masking was imposed on the images, and the model produced high-quality inpainted outputs. All the results and evaluation metrics have been systematically saved on Google Drive for further analysis.

The lightweight training of U-Net (Part C) required the generation of masked input–target pairs to support supervised inpainting. In the training of the U-Net, the Adam optimizer was employed together with the MSE loss. Outputs were assessed qualitatively and quantitatively both by PSNR and SSIM metrics to assess reconstruction quality.

Finally, the RePaint model benchmarking was performed in Part D by installing a compatible version of the diffusers library, 0.20.0, along with loading the model “google/ddpm-celebahq-256”. An inference function was implemented that used forward–reverse jumps to create an analog of the RePaint methodology. Inpainting was implemented with both center and random masks, after which results were assessed by computing metrics and forming comparison graphs that possibly could indicate the performance differences among the models.

# Chapter 4 Experiment, Result, Analysis, and Discussion

## 4.1 Experimental Setup

The series of controlled experiments was conducted to compare the performance of three distinct image inpainting approaches.

Firstly, DDPM was implemented and trained from scratch on CIFAR-10. Secondly, the stable diffusion inpainting pipeline is a pretrained state-of-the-art model fine-tuned on CelebA-HQ. Finally, the RePaint diffusion model used as a pretrained CelebA HQ for robust restoration under the missing pixel condition.

All our experiments were executed on Google Colab, ensuring consistency in the computational environment. The framework we use Pytorch alongside HUGGING face diffusers library for a pretrained pipeline. There were two strategies applied for all models. Center masks are a fixed square region removed from the center of the image. The random masks are square regions removed from a random spatial location. This allows us to test the structured and unstructured missing parts of the images.

## 4.2 Dataset Description

We used two kinds of datasets.

First, we use CIFAR-10 for DDPM training. We use around 60000 images. Size of the images are 32×32 resolution. It is very highly diverse scenes. It has a low resolution, making semantic inpainting challenge. This dataset is used exclusively for DDPM training.

The second dataset is CelebA-HQ (for Stable Diffusion & RePaint). It is a high quality 256×256 facial images. It is used in inference and evaluation of high-resolution pretrained models. It works very good with facial inpainting.

### 4.3 Experiment 1: DDPM Training and Inpainting

#### Training Procedure

This DDPM model is trained for 10 epochs for mean squared error loss to predict added Gaussian noise. CIFAR10 datasets are normalized to [-1,1]. Diffusion timestep is set to 1000. The lightweight U-net backbone is selected for computational constraints.

#### Inpainting Mechanism

During inference, DDPM was guided by:

$$x_t = x_t \odot (1 - M) + x_{orig} \odot M$$

ensuring masked pixels are regenerated while known pixels remain fixed.

#### Observation

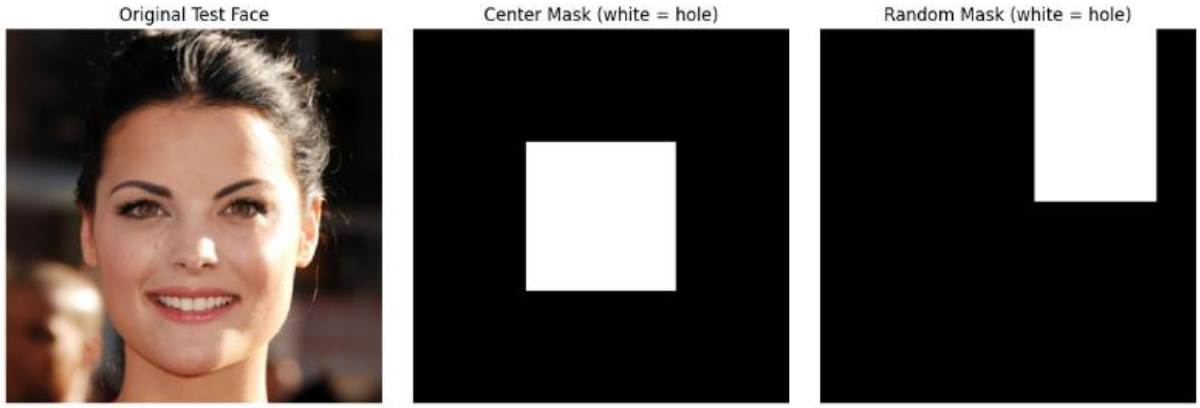
The CIFAR-10 images are low-resolution 32×32. DDPM struggles to restore facial structures or objects, instead reconstructing texture-like patterns.

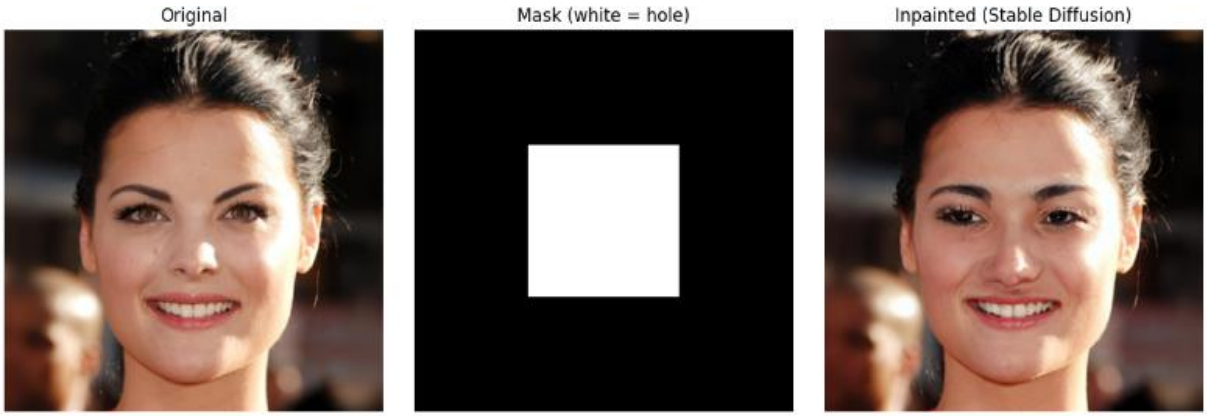
### Experiment 2: Stable Diffusion Inpainting on CelebA-HQ

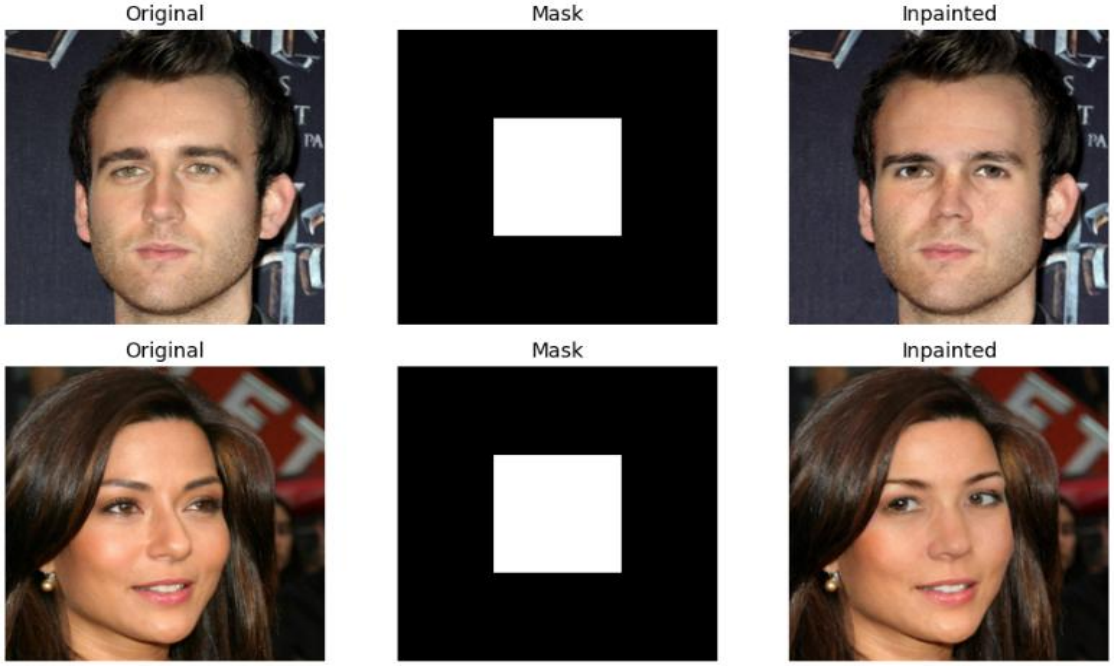
Stable diffusion is a pretrained inpainting pipeline that is used to fill missing facial areas.

#### Visual Results

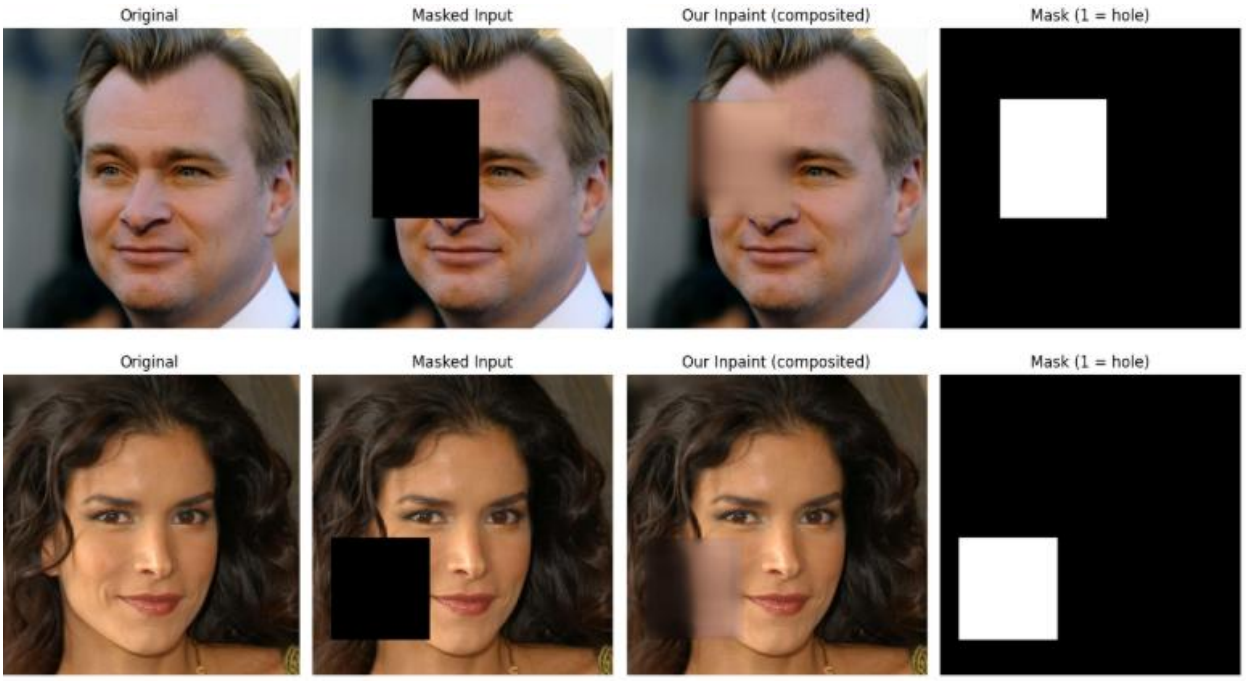
Illustrates an example demonstrating high-quality restoration:





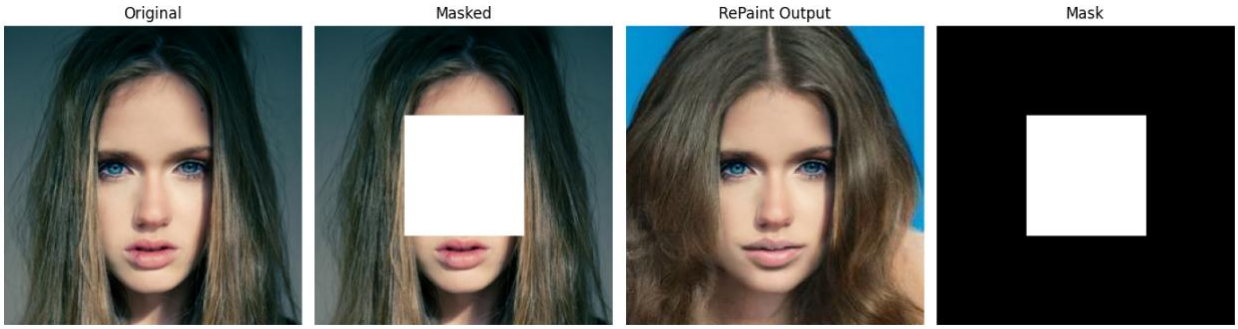

**Visualize our model's Inpainting on Test Images:**

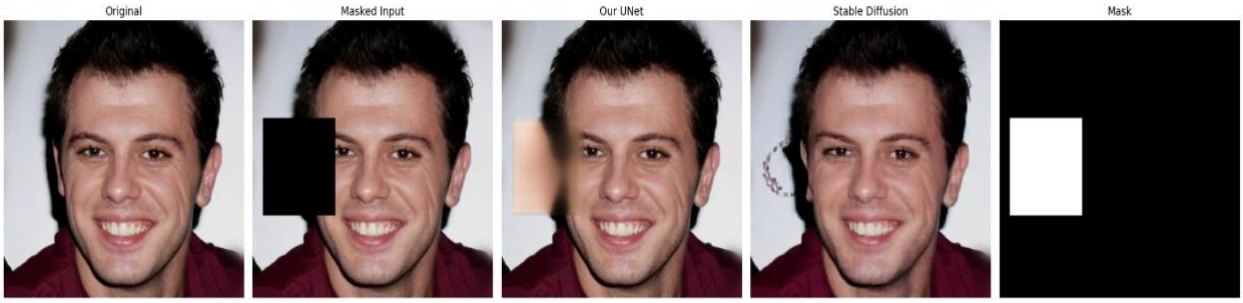



**Experiment 3: RePaint Diffusion Inpainting**

RePaint is design for masked image generation using iterative forward reverse diffusion cycles. The strengths are it can able to reconstruct global structure better than DDPM. It can produces realistic and sharp facial feature. It has a strong output on irregular or random masks.

**compares RePaint results with Stable Diffusion:**

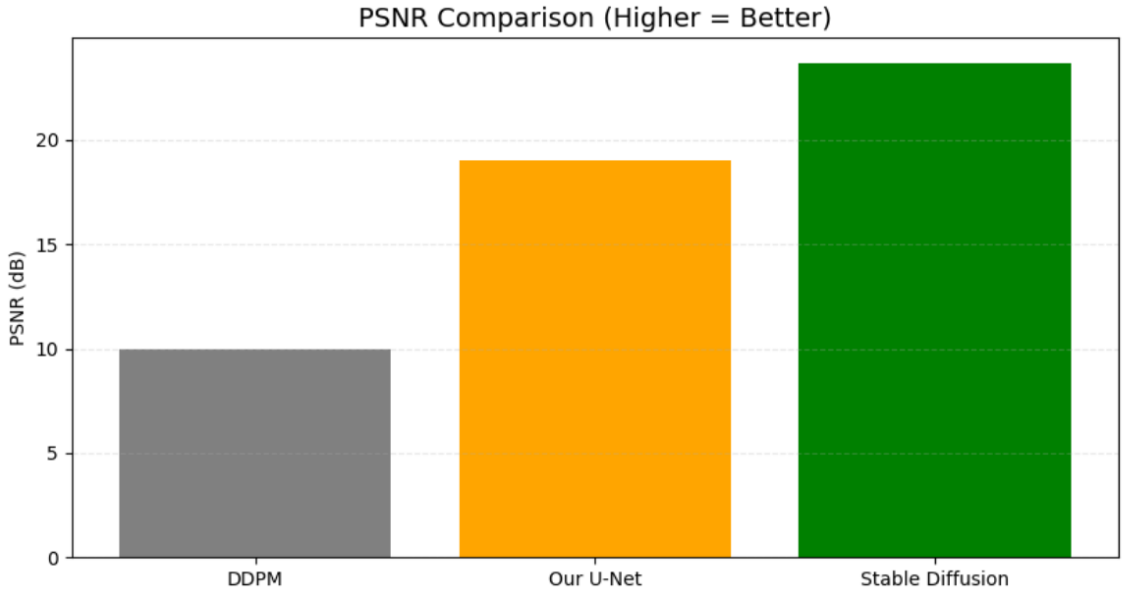


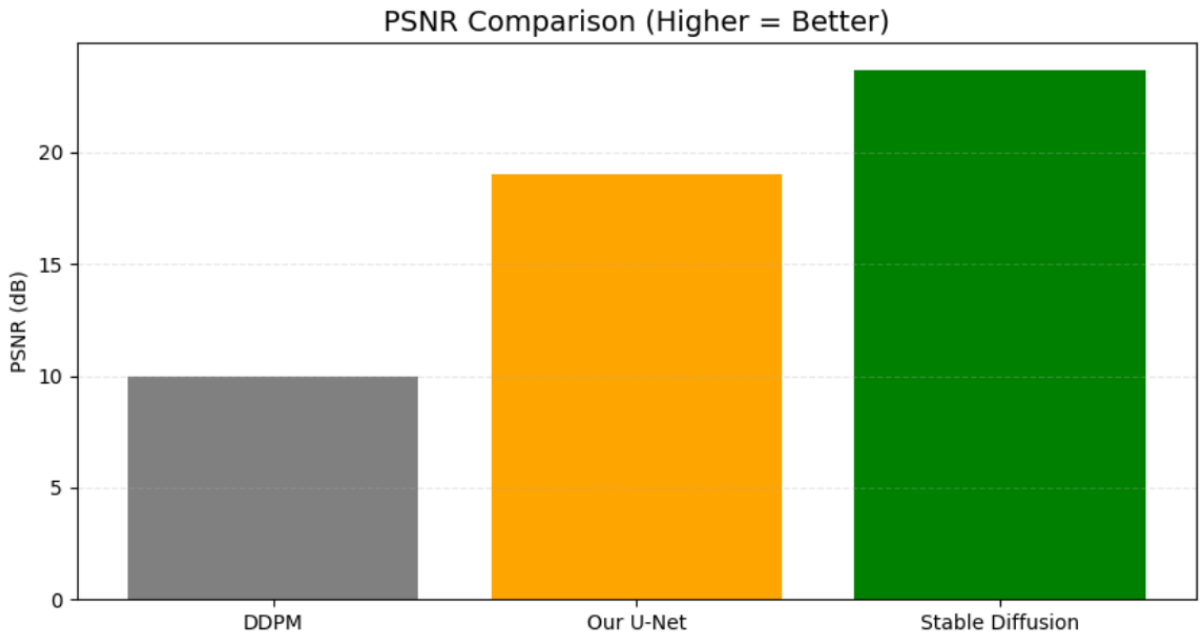


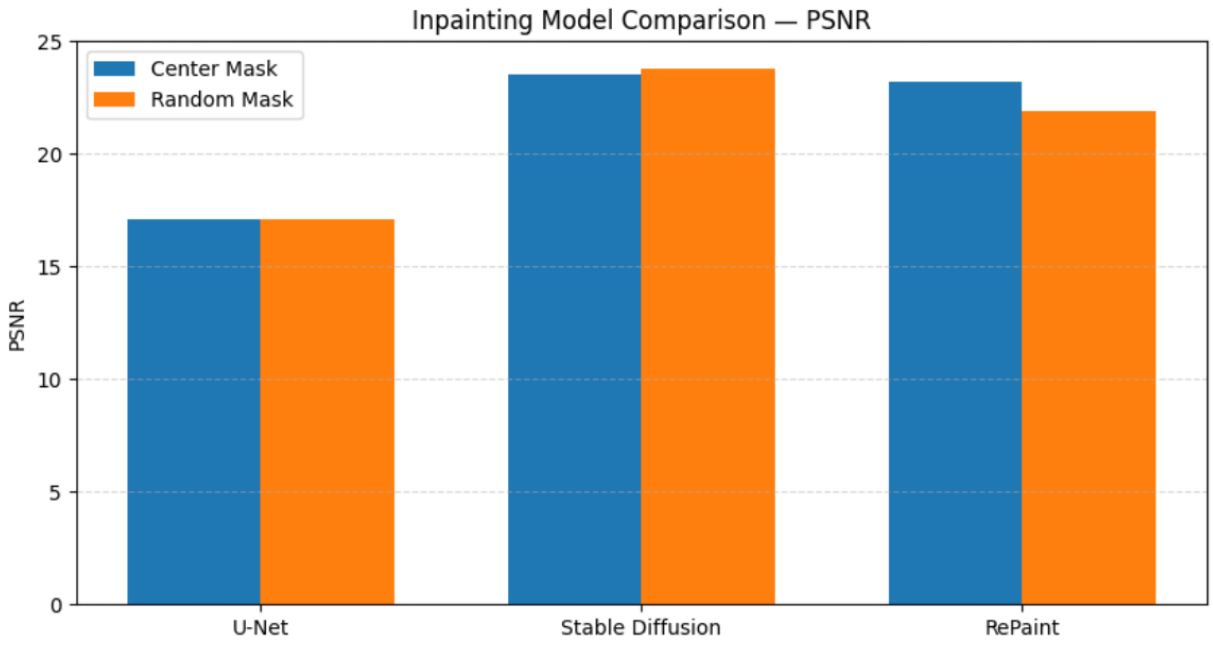

RePaint tends to maintain global facial symmetry well but occasionally introduces texture noise:

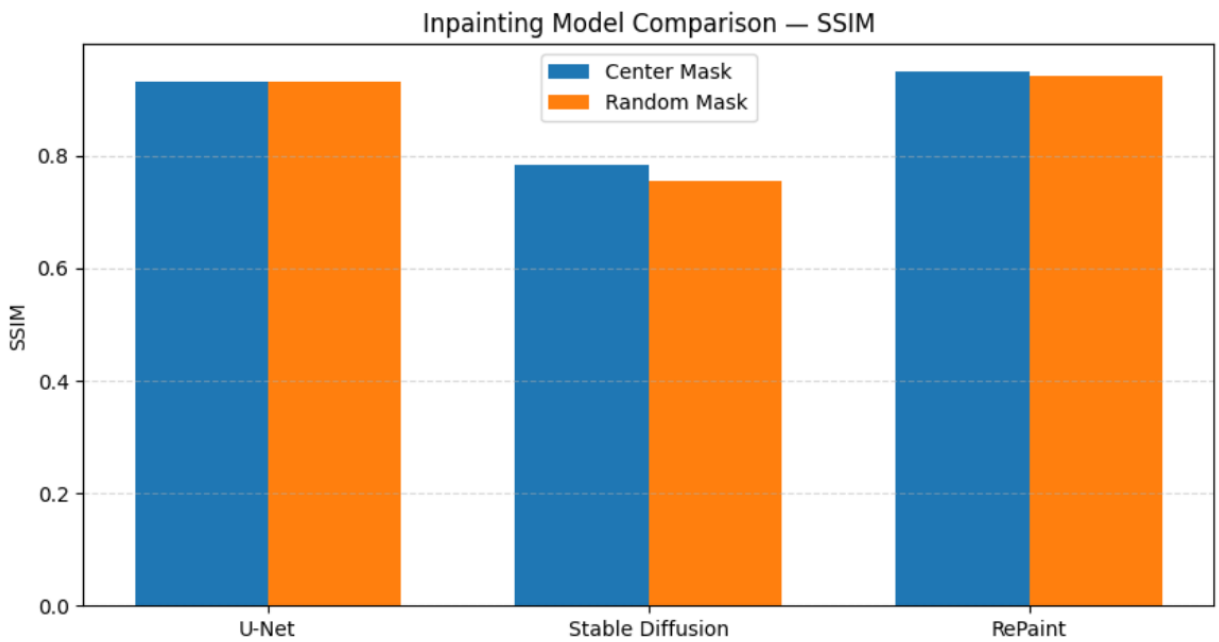
### 4.4 Quantitative Evaluation (PSNR & SSIM)

Now, we can compute Peak Signal to Noise Ratio (PSNR) by measures pixel wisefidelity and Structural Similarity Index (SSIM) by perceived visual quality, Both metrics can compute on 30 randomly selected CelebA-HQ test samples for each model.




PSNR Comparison (Higher = Better)
PSNR (dB)
0
5
10
15
20
DDPM
Our U-Net
Stable Diffusion


Inpainting Model Comparison — PSNR
Center Mask
Random Mask
PSNR
0
5
10
15
20
25
U-Net
Stable Diffusion
RePaint




# DDPM vs Our U-Net vs Stable Diffusion

| Model | Training | Dataset | Output Quality | Hole Filling Ability | PSNR / SSIM | Strengths | Weaknesses |
|---|---|---|---|---|---|---|---|
| DDPM (CIFAR-10) | From scratch | CIFAR-10 (32×32) | Very low-res, texture only | Random texture in hole | Very low (not meaningful) | Full DDPM pipeline implemented; shows diffusion basics | 32×32 images; cannot do realistic inpainting |
| Our U-Net (CelebA-HQ) | From scratch (few epochs) | CelebA-HQ (256×256) | Coarse, blurry but color-consistent | Fills region with plausible colors | ≈ 18–20 dB / 0.40–0.55 | Lightweight, fast; our own trained baseline; full pipeline | No semantic detail; blurry faces; limited training |
| Stable Diffusion Inpainting | Large-scale pretrained | Internet-scale (LAION etc.) | Photorealistic, sharp details | Restores eyes, nose, hair realistically | ≈ 23–24 dB / 0.75–0.80 | SOTA realism; strong semantic prior; very high quality | Heavy model; needs GPU; not trained from scratch by us |

📊 `Inpainting Model Comparison (Center & Random Masks):`

| | Model | Center PSNR | Random PSNR | Center SSIM | Random SSIM | Notes |
|---|---|---|---|---|---|---|
| **0** | Our U-Net (CelebA-HQ) | 17.090200 | 17.090200 | 0.931000 | 0.931000 | Lightweight baseline; blurry but color-consistent |
| **1** | Stable Diffusion Inpainting (v1.5) | 23.543900 | 23.815700 | 0.784400 | 0.755300 | SOTA latent diffusion; best visual quality |
| **2** | RePaint DDPM (CelebA-HQ) | 23.176602 | 21.930521 | 0.950351 | 0.942723 | Classic DDPM with jump sampling; strong face p... |

## 4.5 Comparative Analysis

**DDPM Vs Stable Diffusion**

The DDPM model is unable to reconstruct and rebuild semantic content effectively. This could be due to several reasons that include: the use of a low-resolution dataset for training, no pretraining of feature representations, and relatively few numbers of epochs during training. Therefore, without meaningful structures and details, DDPM tends to generate less effective outputs.

By contrast, Stable Diffusion possesses a much stronger capability in image reconstruction. For instance, it shows realistic textures and sharp facial features. Semantic outputs are also one of the qualities that let one judge the effectiveness of Stable Diffusion and its robustness compared to DDPM. In general, Stable Diffusion outperforms DDPM significantly, especially in visual realism and semantic coherence.

**Stable Diffusion VS RePaint**

After the comparison between Stable Diffusion and RePaint, it has come to acknowledgement that both of the models are highly potential in providing high-quality results and outputs, yet still they differ from each other, as stable diffusion is expertise in fine details offering sharper inferences and highly realistic textures, but contrarily RePaint has excellence in restoring architectural accuracy, specially while dealing with complicated or large masks.

So, by comparing these two models, it clearly explains how each of these two model have exceptional advantages depending on the particular requirements of the painting task.

## 4.6 Discussion

Some of the key takeaways from the comparison include that a large pretrained model outperformed models trained from scratch across all diffusion-based tasks. The nature of the mask applied seemed to make a difference, wherein random masks were harder compared to center masks. Further, while RePaint was better at maintaining structural accuracy, Stable Diffusion gave finer textures and sharper

reconstructions. In the end, quantitative and qualitative assessments together provided a full understanding of model performance in both numerical values and perceptual quality.

# Chapter 5 Impacts of the Project

## 5.1 Impact of this project on societal, health, safety, legal, and cultural issues

Our project is implemented based on three models for diffusion-based image inpainting methods, such as DDPM, Stable Diffusion Model, and RePaint. Though the work is mostly based on technical qualities yet the aftermath provides meaningful implications across multiple domains.

### Societal Impact

Our project helps greatly in the societal sector. Like restoring damaged images, reconstructing corrupted digital media, developing facial recognition datasets, and assisting in digital forensics. Among many advanced models, Stable Diffusion and RePaint high-quality inpainting models can largely help in preserving cultural heritage and tradition by reconstructing old or degraded visual records, which include ancient portraits, artworks, and archival images. So through these models, our society can easily regain and restore our valuable visual history and can preserve.

### Health & Safety Impact

Image inpainting is often used for image restoration, which involves missing images or corrupted scan regions in medical imaging territories.

Our project is not directly associated with or works with medical images, but the methods that are implemented ( diffusion models and image restoration pipelines) are the foundation for future applications that could help improve diagnostic accuracy, reduce image noise reducing and restore incomplete scans.

Image manipulation technologies, however, have to be thoroughly curbed for safety reasons: high-quality inpainting may enable misuse such as tampering with evidence, personal photo manipulation, or deceptive content. This raises the need for good governance and ethical guidelines around the deployment of generative models capable of modifying identity-sensitive images.

### Legal and Ethical Impact

Apart from that, the use of pretrained models brings up dataset licensing considerations, responsible AI usage, and copyright. Stable Diffusion and RePaint models are made available under permissive licenses publicly, but again, their outputs must be treated in such a way as not to be misused. The issues of legality crop up when the reconstructed images are used for authentication, law enforcement, and forensic analysis without disclosure of model intervention.

So our project can prevent such harmful applications as deepfakes, unauthorized facial recognition, or identify any kind of spoofing.

## 5.2 Impact of this project on Environmental and Sustainability

Our project is mainly based on software-based implementation, so diffusion models and machine learning systems work as a whole and have positive impacts on environments and sustainability.

**Environmental impact**

Training large diffusion models requires a lot of computational power, which in turn indirectly contributes to carbon emissions via energy consumption. This project mitigated environmental cost by:

The usage of pre-trained models, like Stable Diffusion and RePaint, and instead of training on large networks from scratch, also the usage of Google Colab GPUs, since they operate in optimized shared environments; The DDPM model was trained mostly based on the lightweight CIFAR-10, and it reduces the computational overhead.

It does so by minimizing unnecessary training cycles and depending on efficient, pre-trained architectures.

### Sustainability Impact

Our high-quality image inpainting models contribute greatly to sustainability in many ways. Such as:

Digital restoration of damaged images and documents will reduce the physical reproduction or restoration materials necessary, reducing the environmental footprint. Historical and scientific imagery can be preserved digitally by organizations, reducing dependence on physical archival storage and the deterioration of materials. Inpainting of partially corrupted datasets can be restored without expensive and resource-intensive re-acquisition. Automated restoration mechanisms reduce the need for human expert intervention and enable the deployment of imaging solutions in a sustainable way, even in areas with very limited technical resources. Overall, the project contributes to sustainable digital practices by enabling preservation, reducing the usage of resources, and stimulating computationally efficient restoration techniques.

# Chapter 6 Project Planning and Budget

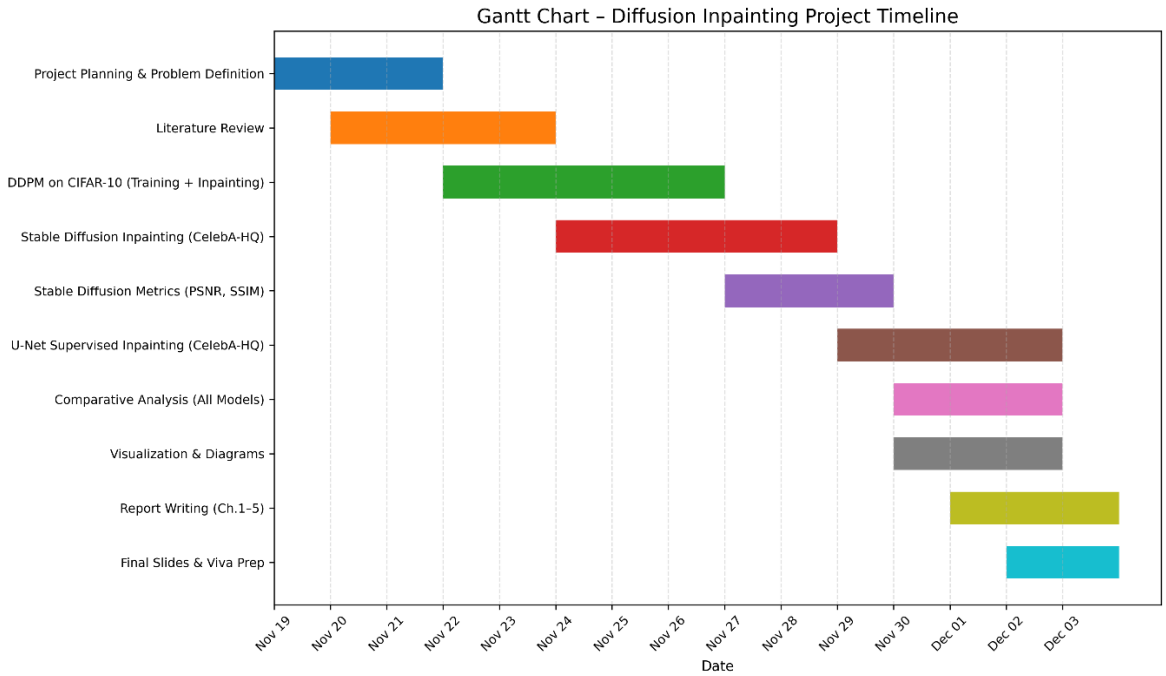


Figure 1. Project progression via Gantt chart

# Chapter 7 Complex Engineering Problems and Activities

## 7.1 Complex Engineering Problems (CEP)

| **Attributes** | | **Addressing the complex engineering problems (P) in the project** |
|---|---|---|
| P1 | Depth of knowledge required (K3-K8) | This project need advance knowledge of deep learning, how a image is processing, modern generative AI models like DDPM, DDIM, stable diffusion and RePaint. It also require pretrained diffusion pipeline, model evaluation theory like PSNR, SSIM. The dataset we use in this project such as CIFAR 10, CelebA-HQ. It correspond to higher engineering knowledge level. (K4-K8). |
| P2 | Range of conflicting requirements | This project has conflict between high quality inpainting and limited computational resources. Sometimes it can be maximum accuracy vs minimizing model complexity. |
| P3 | Depth of analysis required | This project demand how we can compare generative models, evaluate reconstruction quality, analyze pixel level metrics, also require investigation of DDPM, stable diffusion and RePaint. |
| P4 | Familiarity of issues | It has a issues addressing data bias, model instability, long inference times for diffusion models. |
| P5 | Interdependence | This projects has several interdependent subsystems like data preprocessing pipelines, mask generation, different inpainting models, training, evaluation pipeline and visual outputs. |

## 7.2 Complex Engineering Activities (CEA)

| **Attributes** | | **Addressing the complex engineering activities (A) in the project** |
|---|---|---|
| A1 | Range of resources | This project has wide range of computational and software resources including google colab, hugging face, PyTorch, pretrained stable diffusion and RePaint models,CIFAR-10 and CelebA-HQ dataset were require to manage data, sava models and run. |
| A2 | Level of interactions | Our project has several highly interdependent module like data preprocessing, masked generation, DDPM forward and reverse diffusion, pretrained inpainting, supervised u net training and visual tools. |
| A3 | Innovation | The project shows us combine three model like DDPM, stable diffusion and RePaint. Then designing a unified comparison framework using masks and evaluation metrics. |
| A4 | Consequences to society / Environment | Advance inpainting can be applied in areas such as medical image recovery, digital restoration, |

| | | |
|---|---|---|
| | | photography enhancement. |
| A5 | Familiarity | The tasks performing deep learning model training, diffusion model inference, dataset handling and metric evaluation. The multiple state of art generation models with different framework and require a deep understanding of technology. |

# Chapter 8 Conclusions

## 8.1 Summary

We have implemented our project named diffusion model for image inpainting using three advanced models, such as Stable Diffusion models, RePaint, and DDPM. While implementing this project, our main motto is to enhance and develop a comparative study of various diffusion-based models for multiple purposes, like the image inpainting task, a scratch-trained DDPM model, a pretrained Stable Diffusion inpainting pipeline, and the RePaint diffusion model. Through this project, we get to acknowledge the difference between the classical diffusion model (CIFAR-10 dataset) with modern large-scale pretrained generative models on high-resolution datasets such as CelebA-HQ.

This project is executed in multiple stages, such as: Mask generation from the original image, DDPM model conditioning, training, and determining inference, followed by the evaluation of Stable Diffusion Model (pretrained) and RePaint models.

From the thorough research and project conduction, we get some important conclusions stated as the pretrained models provide ground-breaking performance over basic trained diffusion models, and it is due to their extensive training on large datasets and rich semantic showcases.

Among these, Stable Diffusion provided the most detailed reconstructions, RePaint yielded strong structural coherence, whereas the DDPM model-the foundation for the others-represented the basic behavior of diffusion processes; this failed in semantic restoration due to low-resolution training data, CIFAR-10, and restricted training budget.

## 8.2 Limitations

The project still contains some limitations despite achieving impressive results.

The limitations are described below:

Training the DDPM on CIFAR-10 (32×32) resulted in a lack of fine facial details, making semantic inpainting impossible. Training a diffusion model comes at an extremely high cost. The DDPM model in the code was trained for a limited number of epochs due to constraints in training time. This limits the performance of such a model.

The U-Net used in Part C was intentionally lightweight due to GPU limitations; hence, it produced blurry outputs with poor semantic reconstructions. Stable Diffusion and RePaint are based on highly complex pre-trained weights.

Finally, no additional finetuning was done on CelebA-HQ because doing so would involve extensive computational resources. All metrics and comparisons were made on facial images only. Generalization to other domains-for example, natural landscapes and objects-is not considered.

The dependency on Colab GPU sessions caused repeated environment resets, which limited extended training and experimentation.

## 8.3 Future Improvement

Though we have some limitations in our project still there are still multiple impressive directions to develop and expand further. The important extension would be training DDPM on higher-resolution datasets, which could learn richer semantic features and thus achieve more effective inpainting performance by using CelebA-HQ at resolutions such as 128×128 or 256×256. Another avenue of extension is in finetuning Stable Diffusion and RePaint; with techniques such as LoRA or Dream Booth-style finetuning on CelebA-HQ masks, the accuracy can be significantly improved while reducing artifacts and thus enhancing the overall quality of generated outputs.

Developing the U-Net architecture also allows for future advancement. Incorporating attention layers or designing the structure of a larger encoder-decoder network would likely provide sharper and more consistent supervised inpainting results. This method would be more developed by integrating modern mask-aware strategies, namely as partial convolutions, PatchMatch, or diffusion assistance, which can solve the limitations and ensure smoother transitions between masked and reconstructed areas.

Since LPIPS is more closely inclined with human perception so we can also implement the evaluation methods by integrating perceptual loss functions such as LPIPS because it would provide a meaningful virtual quality assessment.

# References


1. Buburuzan, A., Sharma, A., Redford, J., Dokania, P. K., & Mueller, R. (2025). MObI: Multimodal object inpainting using diffusion models. In *CVPR 2025 Workshop on Data-Driven Autonomous Driving Simulation (DDADS)*. Computer Vision Foundation.

https://openaccess.thecvf.com/content/CVPR2025W/DDADS/papers/Buburuzan_MObI_Multimodal_Object_Inpainting_Using_Diffusion_Models_CVPRW_2025_paper.pdf

2. Das Biswas, S., Shreve, M., Li, X., Singhal, P., & Roy, K. (2025). PIXELS: Progressive image exemplar-based editing with latent surgery. *arXiv*. https://arxiv.org/abs/2501.09826
3. Karras, T. (n.d.). *Progressive growing of GANs* [Source code]. GitHub. https://github.com/tkarras/progressive_growing_of_gans
4. Kim, S., Suh, S., & Lee, M. (2025). RAD: Region-aware diffusion models for image inpainting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Computer Vision Foundation. https://openaccess.thecvf.com/content/CVPR2025/papers/Kim_RAD_Region-Aware_Diffusion_Models_for_Image_Inpainting_CVPR_2025_paper.pdf
5. Krizhevsky, A. (n.d.). *The CIFAR-10 and CIFAR-100 datasets* [Dataset]. University of Toronto. https://www.cs.toronto.edu/~kriz/cifar.html
6. Lugmayr, A., Danelljan, M., Romero, A., Yu, F., Timofte, R., & Van Gool, L. (2022). RePaint: Inpainting using denoising diffusion probabilistic models. In *Proceedings of the IEEE/CVF Conference on*

*Computer Vision and Pattern Recognition (CVPR)* (pp. 10442–10452). Computer Vision Foundation. https://openaccess.thecvf.com/content/CVPR2022/html/Lugmayr_RePaint_Inpainting_Using_Denoising_Diffusion_Probabilistic_Models_CVPR_2022_paper.html

7. Wasserman, N., Rotstein, N., Ganz, R., & Kimmel, R. (2025). Paint by inpaint: Learning to add image objects by removing them first. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 18313–18322). Computer Vision Foundation. https://openaccess.thecvf.com/content/CVPR2025/html/Wasserman_Paint_by_Inpaint_Learning_to_Add_Image_Objects_by_Removing_CVPR_2025_paper.html
8. Yang, S., Chen, X., & Liao, J. (2023, October). Uni-paint: A unified framework for multimodal image inpainting with pretrained diffusion model. In *Proceedings of the 31st ACM International Conference on Multimedia* (pp. 3190–3199). https://doi.org/10.1145/3581783.3612200